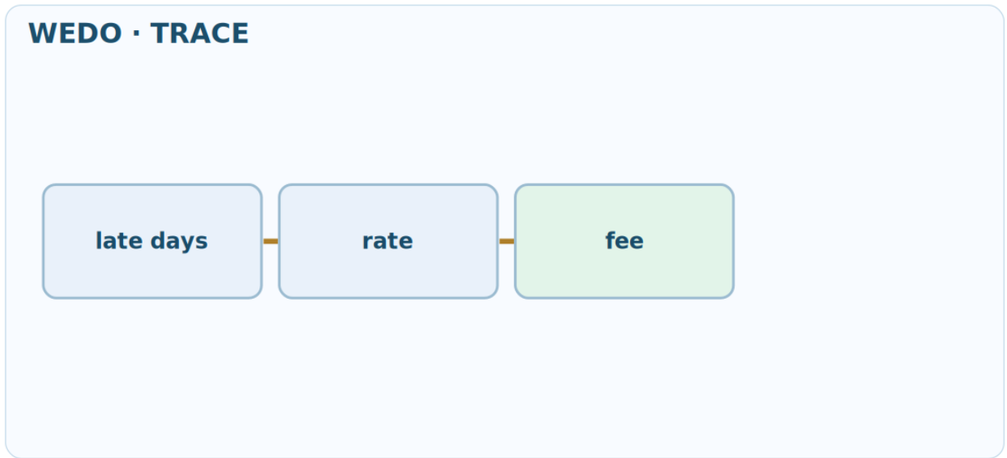
We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Model a library overdue fee of 20p per late day; decide which real-world details matter.

Fee=0.20×lateDays in this simplified rule; real libraries might cap fees or exempt users.



WEDO · TRACE: late days; rate; fee

DO NOW — RECALL

Lesson 059 · Thinking Abstractly: What's the Model?

Name: _____ Class: _____ Date: _____

1. What makes a good field in a record?

2. Why give a unique record key?

3. What is a precondition?

4. Trace binary search for 4 in the sorted, zero-indexed array [2,4,6,8,10]. Start with low=0 and high=4. Use mid=(low+high) DIV 2, high=mid-1 or low=mid+1. State the indexes checked, in order.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Information needed for the model/use case.

2. Model answer

To identify instances unambiguously.

3. Model answer

Assumption that must hold for a calculation to be valid.

4. Model answer

2, 0, 1. Values checked are 6, 2, 4. The search finds 4 at index 1.

By the end, you can…

• Abstraction selects only the information required for the task and deliberately excludes irrelevant real-world detail.

• An abstract model is a simplification of reality: here distance and child/adult status determine the fare; hair colour does not.

• The model is only as good as its assumptions. If peak pricing or concessions apply, the abstraction needs refinement.

Vocabulary — know these terms

abstraction — Focus on relevant details.

model — Simplified representation of reality.

assumption — Condition accepted within a model.

irrelevant — Detail that does not affect task.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Explain
Fee=0.20×lateDays in this simplified rule; real libraries might cap fees or exempt users.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

An abstraction must include every observable detail of reality.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

NEW LEARNING

Read, annotate and explain

Understand

Abstraction selects only the information required for the task and deliberately excludes irrelevant real-world detail.

Model

An abstract model is a simplification of reality: here distance and child/adult status determine the fare; hair colour does not.

Evaluate

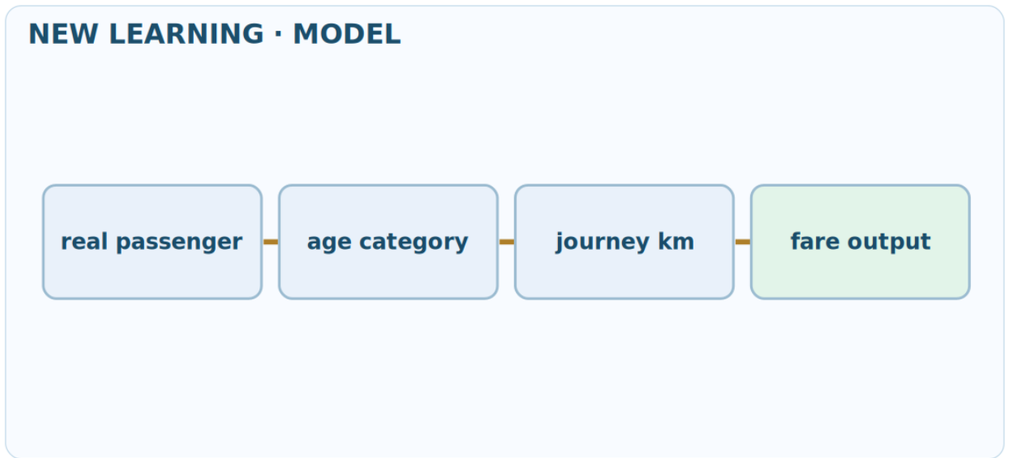
The model is only as good as its assumptions. If peak pricing or concessions apply, the abstraction needs refinement.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



NEW LEARNING · MODEL: real passenger; age category; journey km; fare output

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Evaluate the risks of abstracting a school attendance intervention down to attendance percentage alone. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

For an automated bus fare model, identify three relevant and two irrelevant details. [5]

Question 2

Define inputs, processing and output for library overdue-fee model at 20p/day capped at £2. [6]

I DO — WORKED EXAMPLE

Follow the worked method

Choose relevant inputs for calculating parking fee at £2 per hour, rounded to whole hours.

Explain

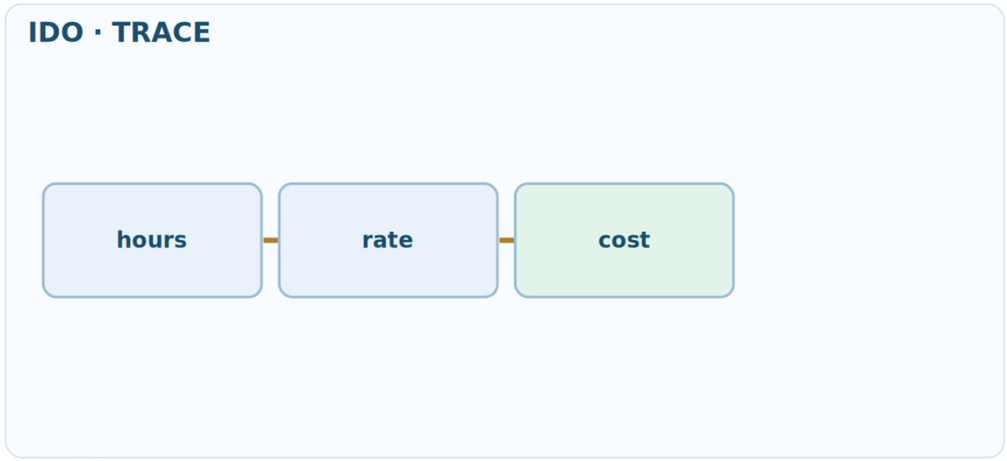
ParkingFee(hours)=hours×£2 for the simplified model; define non-negative hours and rounding rules.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Choose relevant inputs for calculating parking fee at £2 per hour, rounded to whole hours.

ParkingFee(hours)=hours×£2 for the simplified model; define non-negative hours and rounding rules.



IDO · TRACE: hours; rate; cost

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Model a library overdue fee of 20p per late day; decide which real-world details matter.
Source code / input:
Input lateDays (non-negative); rate £0.20/day; output fee. Book cover colour is irrelevant.